

ImageGen2.py ×

```
1 import os
2 import openai
3 import secret
4 from PIL import Image
5 from io import BytesIO
6 import requests
7
8 openai.api_key = secret.api_key
9 # WRITE YOUR CODE HERE
```



Terminal ×

```
Welcome to Ubuntu 18.04.5 LTS (GNU/Linux 5.19.0-1028-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

 * Canonical Livepatch is available for installation.
   - Reduce system reboots and improve kernel security. Activate at:
     https://ubuntu.com/livepatch
 *
 * Welcome to the Codio Terminal!
 *
 * https://docs.codio.com/develop/develop/ide/boxes/overview
 *
 * Your Codio Box domain is: othellojoker-touristsweet.codio.io
 *
Last login: Mon Feb 27 20:05:42 2023 from 192.168.11.179
codio@othellojoker-touristsweet:~/workspace$ ls
exerc.py  ImageGen2.py  imageGen.py  __pycache__  secret.py  setting.py  test.py
codio@othellojoker-touristsweet:~/workspace$
```

Guide ×

Collapse

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Edges

In the context of image processing, "**edges**" refer to boundaries or transitions between different regions in an image. These boundaries usually occur at points where there is a sharp change in color or intensity values. Identifying edges is a crucial step in many computer vision and image processing tasks, as it helps in understanding the structure and objects present in an image. In the next example, we are using filters that focus on identifying and enhancing the edges in the image.

Edge

The edge enhance filter highlights the edges of objects in the image, making them more distinct and pronounced. This filter can be used to create a more stylized or artistic effect.

```
from PIL import ImageFilter

# Apply an edge enhance filter to the base image
edge_enhanced_image = apple_image.filter(ImageFilter.EDGE_ENHANCE)

# Save the edge enhanced image to a file
edge_enhanced_image.save('edge_enhanced_red_apple.jpg')
```

TRY IT!

[Click here to refresh your base apple image](#)
[Click here to refresh your edge_enhanced apple image](#)

Find Edges

The find edges filter is another type of image filter that enhances the edges in an image, similar to the contour filter. However, the find edges filter typically produces a more pronounced and higher contrast effect, making the edges stand out even more.

```
from PIL import ImageFilter

# Apply a find edges filter to the base image
edge_image = apple_image.filter(ImageFilter.FIND_EDGES)

# Save the edge image to a file
edge_image.save('edge_red_apple.jpg')
```

TRY IT!

[Click here to refresh your find edges apple image](#)

In this lesson, we explored various image filters provided by the PIL library and learned how to apply them to images generated by the DALL-E 2 API. By combining these filters with the DALL-E 2 API, you can create a wide array of unique and visually striking images.

Feel free to experiment with different filters and combinations to create your own custom images using the DALL-E 2 API and the PIL library.

What is the purpose of the edge enhance filter?

☐ To soften the image by reducing noise and details

☒ To highlight the edges of objects in the image, making them more distinct and pronounced

☐ To create a high-contrast, black-and-white image

☐ To trace the edges of objects in the image

The edge enhance filter is used to highlight the edges of objects in the image, making them more distinct and pronounced. This filter can be used to create a more stylized or artistic effect.

Check It!

Next